

Problem

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30Days of Code

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Welcome to Day 18! Today we're learning about Stacks and Queues. Check out the [Tutorial](#) tab for learning materials and an instructional video!

A palindrome is a word, phrase, number, or other sequence of characters which reads the same backwards and forwards. Can you determine if a given string, *s*, is a palindrome?

To solve this challenge, we must first take each character in *s*, enqueue it in a queue, and also push that same character onto a stack. Once that's done, we must dequeue the first character from the queue and pop the top character off the stack, then compare the two characters to see if they are the same; as long as the characters match, we continue dequeuing, popping, and comparing each character until our containers are empty (a non-match means *s* isn't a palindrome).

Write the following declarations and implementations:

1. Two instance variables: one for your *stack*, and one for your *queue*.

2. A void `pushCharacter(char ch)` method that pushes a character onto a stack.

3. A void `enqueueCharacter(char ch)` method that enqueues a character in the *queue* instance variable.

4. A char `popCharacter()` method that pops and returns the character at the top of the *stack* instance variable.

5. A char `dequeueCharacter()` method that dequeues and returns the first character in the *queue* instance variable.

Input Format

You do not need to read anything from `stdin`. The locked stub code in your editor reads a single line containing string *s*. It then calls the methods specified above to pass each character to your instance variables.

Constraints

- s* is composed of lowercase English letters.

You have earned 30.00 points!

You are now 3 challenges away from the 4th star for your 30 days of code badge.

57%

19/22

Congratulations

You solved this challenge. Would you like to challenge your friends?

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Next Challenge

Test case 0

Compiler Message

Success

Test case 1

Test case 2

Test case 3

Test case 4

Test case 5

Test case 6

Input (stdin)

Download

1

racecar

Expected Output

Download

1

The word, racecar, is a palindrome.

Buscar

6:30 p. m.

3/8/2026